

# Q-value Regularized Transformer:

## A Faithful Reproduction and a Counterfactual Necessity Extension

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### 1. Introduction

Offline reinforcement learning (ORL) trains a policy from a fixed dataset of past trajectories without further environment interaction. Two paradigms dominate: *Decision Transformer* (DT) casts ORL as return-conditioned sequence modeling, giving stable supervised training but only trajectory-level return-to-go (RTG) signals; *Q-learning* provides state-action-level policy improvement but suffers from bootstrap error, especially in long-horizon, sparse-reward settings. **Q-value Regularized Transformer (QT)** (Hu et al., ICML 2024) combines the two: a DT-style policy is trained jointly with a learned double-Q critic, and at inference 50 candidate actions are drawn from RTG-jittered prompts and re-ranked by the critic. This report covers three phases of our work: (i) a faithful reproduction of QT on two D4RL locomotion tasks, (ii) an attempted extension that injected a supervised reverse-credit signal into the critic — which *failed*, and (iii) the resulting pivot to a different question: whether a learnable *counterfactual necessity* score is a useful new signal alongside Q-value.

### 2. Chosen Result

Our headline reproduction target is the first two rows of Table 1 in Hu et al.: D4RL normalized scores of  $98.5 \pm 1.1$  on `walker2d-medium-replay-v2` and  $102.0 \pm 0.2$  on `hopper-medium-replay-v2`. This is the central evidence that QT outperforms both pure DT and pure Q-learning, and reproducing it exercises the paper’s most non-trivial design choice — 50-way RTG candidate expansion with Q-bootstrap correction at inference — rather than just the base transformer architecture.

### 3. Methodology

#### 3.1 Baseline reproduction (TranQil)

We re-implemented QT from scratch in PyTorch. The policy is a 4-layer, 4-head, 256-dim GPT-2-style causal transformer; the critic is a dual MLP with Mish activations and an EMA target network ( $\tau=0.995$ ). The joint objective is

$$\mathcal{L}_{\text{QT}}(\theta) = \mathcal{L}_{\text{DT}}(\theta) - \alpha \cdot \mathbb{E}[Q_\phi(s_i, \pi_\theta(\tau_t)_i)], \quad \alpha = \eta / \mathbb{E}[|Q_\phi(s, a)|],$$

where  $\mathcal{L}_{\text{DT}}$  is RTG-conditioned behavior cloning and the second term pushes actions toward high learned values; the critic is trained with  $n$ -step Bellman targets under double-Q clipping. At inference, the policy samples 50 candidates from jittered RTGs and Q-softmax-selects. Three implementation choices silently absent from the paper text were essential: GPT-2 positional embeddings must be zeroed and frozen (forcing reliance on timestep embeddings), the Tanh action head must *not* be clipped to `env.action_space`, and evaluation must use `max.steps=1000` — at the gym default of 500, the maximum achievable normalized score is  $\approx 44$ , the most common silent replication failure. Training used D4RL `medium-replay-v2` on a single NVIDIA RTX 5090 ( $\sim 10$  GPU-hours total).

#### 3.2 Failed extension: supervised reverse-credit (SARD-QT)

QT inherits TD bootstrap error from its critic, which is most damaging in long-horizon sparse-reward tasks where reward-relevant actions occur many steps before the first nonzero reward. We hypothesised that a *supervised* signal — “how many steps until the first reward?” extracted retroactively from successful trajectories — could correct the critic where TD signal is weakest. We trained an auxiliary reverse-credit critic on this signal and fused it with the QT critic via several strategies (additive sum, gated weighted average driven by QT-critic disagreement, distillation, and uncertainty-weighted ensembling). All variants exhibited the same failure mode: either the auxiliary critic dominated and collapsed the policy to greedy goal-seeking (eliminating QT’s trajectory-stitching benefit) or it was effectively ignored. None matched our QT baseline on `walker2d/hopper`, and on the sparse-reward `antmaze-medium-diverse` where the supervised signal was supposed to help, both QT and every SARD-QT variant scored 0%. **Diagnostic:** the supervised reverse signal occupies the *same role* as Q-value (state-action value supervision), so fusing them at best yields a noisier ensemble of the same kind of estimate.

### 3.3 Pivot: counterfactual necessity (NCA-RL)

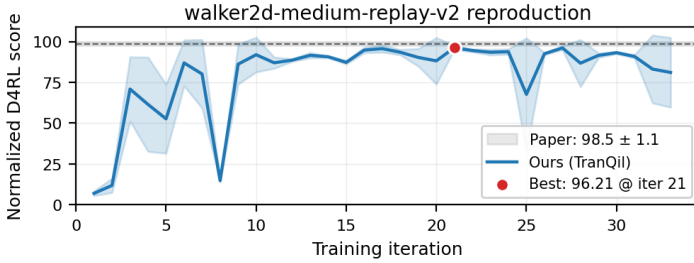
The SARD-QT failure motivated a different question — not how to improve the value signal but how to introduce a *different* signal. We define an action’s *necessity* as the counterfactual change in outcome under an alternative drawn from a perturbation kernel  $\kappa_\varepsilon$ . The ground-truth interventional version is  $\mathcal{N}_\kappa^\pi(s, a) := G_{\text{base}}(s, a) - \mathbb{E}_{\tilde{a} \sim \kappa_\varepsilon}[G_{\text{cf}}(s, \tilde{a})]$  via paired rollouts from the same restored simulator state. The offline proxy is a Q-margin over  $\kappa_\varepsilon$ :

$$\mathcal{N}^{\text{off}}(s, a) := Q_\phi(s, a) - \mathbb{E}_{\tilde{a} \sim \kappa_\varepsilon}[Q_\phi(s, \tilde{a})],$$

and a small two-head MLP ensemble  $\mathcal{N}_\psi$  regresses on  $\mathcal{N}^{\text{off}}$  atop a frozen offline IQL critic.

## 4. Results & Analysis

### QT reproduction



**Figure 2:** Walker2d-mr reproduction; each point: mean (line)  $\pm$  std (band) over 10 rollout episodes per checkpoint. Gray: paper  $98.5 \pm 1.1$  (3 seeds  $\times$  10). Best 96.21 @ iter 21; best episode 99.72.

Walker2d’s  $-2.3$  gap is within the paper’s  $\pm 1.1$  band; Hopper’s larger  $-2.9$  reflects single-seed variance. Q-loss `eta` and `grad_norm` needed more tuning than the paper suggests.

### SARD-QT extensions

No fusion variant beat the QT baseline on walker2d/hopper; on `antmaze-medium-diverse` both QT and every SARD-QT variant scored 0%. **Diagnostic:** the reverse-credit critic and the Q-critic are interchangeable as state-action value estimators; combining redundant estimators of the same quantity does not address the underlying credit-assignment problem.

### Necessity validation

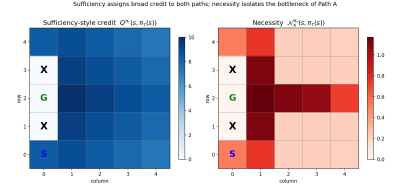
- **Tier-0 (closed-form gridworld, Fig. 1).** Sufficiency contraction 0.85 vs. necessity contraction 0.35: a **2.42 $\times$  separation**. Necessity is well-defined and learnable.
- **Tier-1 (maze2d-large-dense-v2, central finding).**  $\mathcal{N}_\psi$  fits  $\mathcal{N}^{\text{off}}$  well (Spearman  $\rho=0.797$ ); crucially  $\rho(Q_\phi, \mathcal{N}^{\text{off}}) = -0.033$ :

$Q$  and necessity are empirically **orthogonal dimensions**, not transformations of one another — exactly what is required for  $\mathcal{N}$  to be a useful new signal rather than a re-parameterisation of  $Q$ .

- **Tier-1 drop test** (200 paired rollouts via `set_state` restoration).  $\rho(\mathcal{N}_\psi, \Delta G) = +0.164$ ,  $p=0.02$  — significant but weak. The proxy-fit vs. truth-fit gap ( $\rho=0.80$  vs. 0.16) localises the open problem: the Q-margin proxy captures *some* but not most of the true counterfactual signal.

## 5. Reflections

1. **The SARD-QT negative result was diagnostic, not just disappointing.** The long-horizon QT bottleneck is not a missing supervised signal but the conceptual narrowness of value-based supervision itself.
2. **Orthogonality is the design’s success; the proxy-to-truth gap is the open problem.**  $Q \perp \mathcal{N}$  ( $\rho=-0.03$ ) is what makes  $\mathcal{N}$  useful. Future work: swap IQL for calibrated CQL, regress on Monte-Carlo  $\Delta G$ , and close the loop with necessity-weighted online fine-tuning.



**Figure 1:** Tier-0 gridworld: Q-sufficiency (left) credits both paths; necessity (right) isolates the bottleneck.

| Environment | Paper           | Ours  | Gap    |
|-------------|-----------------|-------|--------|
| walker2d-mr | $98.5 \pm 1.1$  | 96.21 | $-2.3$ |
| hopper-mr   | $102.0 \pm 0.2$ | 99.10 | $-2.9$ |

**Table 1:** Normalized D4RL scores on `medium-replay-v2`. Paper: Hu et al. 2024 (3 seeds); ours: 1 seed.